Optimal Control of Twin Rotor MIMO system Using Output Feedback

Bhanu Pratap, *Student Member, IEEE*, Abhishek Agrawal, and Shubhi Purwar

***Abstract*— In this paper an optimal state regulator is designed for the twin rotor multi-input-multi-output (MIMO) system. The twin rotor MIMO system (TRMS) exemplifies a high order nonlinear system with significant cross couplings. From the nonlinear model of TRMS a linearised model is obtained and a controller is designed to regulate the states. The controller gain is updated iteratively, until optimal value is reached. Finally simulation results are presented to show the effectiveness of the proposed controller for the TRMS.**

***Index Terms*-- Nonlinear coupled system, optimal control, output feedback, twin rotor MIMO system.**

I. Introduction

THE modeling and control of the TRMS [1] has gained a lot of attention because the dynamics of the TRMS and a helicopter are similar in certain aspects [2]-[4]. Due to the high coupling effect between two propellers, unstable and nonlinear dynamics, the control problem of the TRMS has been considered as a challenging research topic. In [5] a decoupling control using robust deadbeat control technique is proposed for TRMS. In [6] a PID control is design using genetic algorithms for TRMS. A comparative analysis between classical control and intelligent control for TRMS is given in [7]. Adaptive fuzzy based controllers are designed in [8]-[10] for the tracking of pitch and yaw angles. NN based adaptive nonlinear model inversion control for TRMS is presented in [11]. Feedback linearization controller has been proposed for TRMS in [12], [13]. All state variables are assumed to be measurable which is practically not feasible.

In the laboratory setup of TRMS all the states are not available to measure, and hence if a controller based on state feedback is to be designed, then an observer is needed which will estimate the unmeasurable states of TRMS, as proposed in [14]-[16]. Due to nonlinearity, the observer design cannot be carried out independent of the state feedback design.

In other words, the separation principle is not applicable, unlike linear control theory. In fact, even in linear control design, when model uncertainties are taken into consideration, the design of the observer cannot be separated from the design of state feedback control [17]. In [18] an optimal output-feedback controller for single-input linear systems is proposed. An optimal control technique for weakly coupled nonlinear systems is proposed in [19]. An optimal control technique for TRMS is presented in [20], here the TRMS is decoupled into two independent SISO systems, for horizontal and vertical motions, and separate controllers are designed for the two systems, considering the coupling effects as disturbances.

In this paper an optimal controller for the TRMS is presented, which uses the output feedback, i.e., it is considered that the measures of outputs are available. It avoids the need of knowledge of all the states of the system. Therefore the observer design to estimate the unmeasurable states is avoided in the proposed method.

The remainder of the paper is arranged as follows. In Section II, the TRMS system is introduced and the parameters of the system are specified. In Section III, the problem statement is given. The proposed optimal controller is introduced in Section IV. The controller performance is demonstrated in Section V by providing simulation results on the TRMS. Finally concluding remarks are made in Section VI.

II. 2-DOF TRMS System and Model

The TRMS mechanical unit has two rotors placed on a beam together with a counterbalance (as shown in Fig. 1), whose arm with a weight at its end is fixed to the beam at the pivot and it determines a stable equilibrium position. The beam is pivoted on its base in such a way that it can rotate freely both in the horizontal and vertical planes. Either the horizontal or the vertical degree of freedom can be restricted to 1 degree of freedom (1-DOF) using nylon screws found near pivot point. At both ends of the beam there are rotors (the main and tail rotors) driven by dc motors. The main rotor produces a lifting force allowing the beam to rise vertically making a rotation around the pitch axis. While, the tail rotor is used to make the beam turn left or right around the yaw axis. The whole unit is attached to the tower allowing for safe helicopter control experiments.

Apart from the mechanical units, the electrical unit (placed under the tower) plays an important role for TRMS control. It allows for measured signals transfer to the PC and control signal application via an I/O card. The mechanical and electrical units provide a complete control system setup. This device is a multivariable, nonlinear and strongly coupled

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system, with degrees of freedom on the pitch and yaw angle denoted by ψ and φ respectively. The system parameters of the TRMS are given in Table 1.

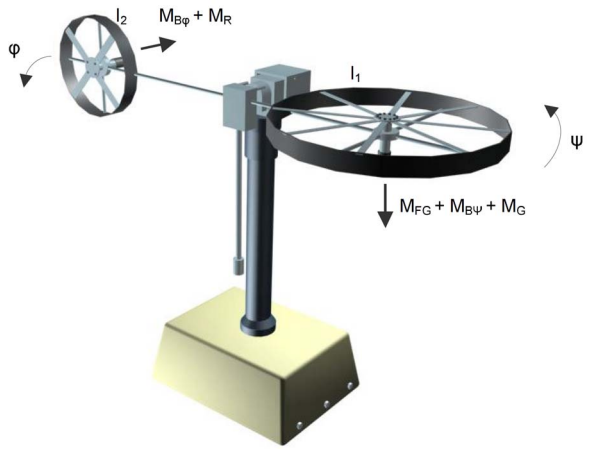


Fig. 1. TRMS Model

The complete dynamics of the TRMS system can be represented in the state-space form as follows.

$$\left.\begin{aligned}
&\frac{d}{dt}\psi = \dot{\psi}\\
&\frac{d}{dt}\dot{\psi} = \frac{a_1}{I_1}\tau_1^2 + \frac{b_1}{I_1}\tau_1 - \frac{M_g}{I_1}\sin(\psi) + \frac{0.0326}{2I_1}\sin(2\psi)\dot{\varphi}^2\\
&\qquad -\frac{B_{1\psi}}{I_1}\dot{\psi} - \frac{k_{gy}}{I_1}\cos(\psi)\dot{\varphi}\left(a_1\tau_1^2 + b_1\tau_1\right)\\
&\frac{d}{dt}\phi = \dot{\phi}\\
&\frac{d}{dt}\dot{\varphi} = \frac{a_2}{I_2}\tau_2^2 + \frac{b_2}{I_2}\tau_2 - \frac{B_{1\varphi}}{I_2}\dot{\varphi} - \frac{k_c}{I_2}1.75\left(a_1\tau_1^2 + b_1\tau_1\right)\\
&\frac{d}{dt}\tau_1 = -\frac{T_{10}}{T_{11}}\tau_1 + \frac{k_1}{T_{11}}u_1\\
&\frac{d}{dt}\tau_2 = -\frac{T_{20}}{T_{21}}\tau_2 + \frac{k_2}{T_{21}}u_2
\end{aligned}\right\} \quad (1)$$

The state and output vectors are given by

$$\begin{aligned}
x &= \begin{bmatrix}\psi & \dot{\psi} & \varphi & \dot{\varphi} & \tau_1 & \tau_2\end{bmatrix}^T\\
y &= \begin{bmatrix}\psi & \varphi\end{bmatrix}^T
\end{aligned} \quad (2)$$

where

ψ : Pitch (elevation) angle

φ : Yaw (azimuth) angle

τ_1 : Momentum of main rotor

τ_2 : Momentum of tail rotor

TABLE 1: TRMS SYSTEM PARAMETERS

Parameters	Values
I_1 = Moment of inertia of vertical rotor	6.8×10^{-2} $kg-m^2$
I_2 = Moment of inertia of horizontal rotor	2×10^{-2} $kg-m^2$
a_1 = Static characteristic parameter	0.0135
b_1 = Static characteristic parameter	0.0924
a_2 = Static characteristic parameter	0.02
b_2 = Static characteristic parameter	0.09
M_g = Gravity momentum	0.32 $N-m$
$B_{1\psi}$ = Friction momentum function parameter	6×10^{-3} $N-m-s/rad$
$B_{1\varphi}$ = Friction momentum function parameter	1×10^{-1} $N-m-s/rad$
k_{gy} = Gyroscopic momentum parameter	0.05 s/rad
k_1 = Motor 1 gain	1.1
k_2 = Motor 2 gain	0.8
T_{11} = Motor 1 denominator parameter	1.1
T_{10} = Motor 1 denominator parameter	1
T_{21} = Motor 2 denominator	1
T_{20} = Motor 1 denominator parameter	1
T_p = Cross reaction momentum parameter	2
T_0 = Cross reaction momentum parameter	3.5
k_c = Cross reaction momentum gain	–0.2

III. PROBLEM STATEMENT

Let the TRMS plant can be represented as the continuous-time linear system given by

$$\begin{aligned}
\dot{x} &= Ax + Bu\\
y &= Cx
\end{aligned} \quad (3)$$

with $x \in \mathbb{R}^6$ as the states, $u \in \mathbb{R}^2$ as the control input and $y \in \mathbb{R}^2$ as the measured output. Where the system matrices can be obtained by linearizing the nonlinear system of (1) as

$$A = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0\\
\frac{-M_g}{I_1} & \frac{-B_{1\psi}}{I_1} & 0 & 0 & \frac{b_1}{I_1} & 0\\
0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & \frac{-B_{1\phi}}{I_2} & -1.75k_c\frac{b_1}{I_2} & \frac{b_2}{I_2}\\
0 & 0 & 0 & 0 & \frac{-T_{10}}{T_{11}} & 0\\
0 & 0 & 0 & 0 & 0 & \frac{-T_{20}}{T_{21}}
\end{bmatrix};$$

$$B = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{k_1}{T_{11}} & 0\\
0 & 0 & 0 & 0 & 0 & \frac{k_2}{T_{21}}
\end{bmatrix}^T;\ C = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0 & 0
\end{bmatrix}$$

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Using the output feedback, the control input will be of the form

$$u = -Ky \tag{4}$$

where $K \in \mathbb{R}^{2\times 2}$ is a matrix of constant feedback coefficient to be determined by the design parameters. Since the regulator problem involves only stabilisation of the plant and inducing good closed loop time response, the u will be taken as a pure feedback with no auxiliary input.

In the regulator problem the interest is in obtaining a good time response as well as the stability of closed loop system, so considering a performance criterion in time domain, given by

$$J = \frac{1}{2}\int_0^\infty \left(x^T Qx + u^T Ru\right) dt \tag{5}$$

where Q is positive semidefinite matrix and R is positive definite matrix (i.e. $Q \geq 0$ and $R > 0$). Here the Q matrix must be chosen such that $\left(Q^{1/2}, A\right)$ is detectable. The definiteness assumption on matrices Q and R, guarantee that the performance index J is non-negative and lead to a sensible minimization problem. We have to find a control input $u(t)$ to minimize the above performance index.

IV. Optimal Controller Design Based on Output-Feedback

The design of linear quadratic regulator with output feedback will be as follows. Given the linear continuous time system (3), find the value of controller gain K in the control input u of (4), such that the performance index J of (5) is minimized [21]. From (3) and (4)

$$u = -KCx \tag{6}$$

from (3) and (6)

$$\dot{x} = (A - BKC)x \equiv A_C x \tag{7}$$

The performance index J can be expressed as

$$J = \frac{1}{2}\int_0^\infty \left\{x^T \left(Q + C^T K^T RKC\right)x\right\} dt \tag{8}$$

The minimization problem of the performance index of (8) is a dynamic optimization, which may be converted into an equivalent static optimization problem, by considering a positive semidefinite matrix P such that

$$\begin{aligned}\frac{d}{dt}\left(x^T Px\right) &= -x^T \left(Q + C^T K^T RKC\right)x \\ &= \dot{x}^T Px + x^T P\dot{x} = x^T \left(A_C^T P + PA_C\right)x\end{aligned} \tag{9}$$

Then the performance index J may be written as

$$J = \frac{1}{2}x^T(0)Px(0) - \frac{1}{2}\lim_{t\to\infty} x^T(t)Px(t) \tag{10}$$

Assuming that the closed loop system is asymptotically stable so that $x(t)$ vanishes with time, then the performance index of (10) can be written as

$$J = \frac{1}{2}x^T(0)Px(0) \tag{11}$$

from (9)

$$A_C^T P + PA_C + C^T K^T RKC + Q = 0 \equiv g \tag{12}$$

This equation is known as Lyapunov equation. For any fixed controller gain matrix K if there exists a constant, symmetric, positive semidefinite matrix P that satisfies (12), and if the closed loop system is stable, then the cost J is given in terms P by (11).

The performance index can also be written as

$$J = \frac{1}{2}tr\left(PX\right) \tag{13}$$

where X is a 6×6 symmetric matrix, defined by

$$X \equiv x(0)x^T(0) \tag{14}$$

Now to solve the optimal control problem defining a Hamiltonian as

$$\mathrm{H} = tr\left(PX\right) + tr\left(gS\right) \tag{15}$$

with S being a 6×6 symmetric matrix of Langrange multipliers. Now our problem of optimization is a problem of minimization of the Hamiltonian [22]. To do this, setting the partial derivatives of H with respect all the independent variables P, S and K; equal to zero.

$$\left.\begin{aligned}\frac{\partial \mathrm{H}}{\partial S} &= A_C^T P + PA_C + C^T K^T RKC + Q = 0 \equiv g \\ \frac{\partial \mathrm{H}}{\partial P} &= A_C S + SA_C^T + X \\ \frac{1}{2}\frac{\partial \mathrm{H}}{\partial K} &= RKCSC^T - B^T PSC^T\end{aligned}\right\} \tag{16}$$

The (16) gives us the necessary condition for the solution of the linear quadratic regulator with output feedback.

The first two equations of (16) will give the values of matrices P & S; and the third equation will give the value of gain K.

$$K = R^{-1}B^T PSC^T \left(CSC^T\right)^{-1} \tag{17}$$

The optimal value of controller gain K will be calculated iteratively, until the difference between the values of performance index J in the two successive iterations becomes less than a very small value, or we can say if they become almost equal, i.e. $J_i - J_{i+1} < \varepsilon$; where ε is a very small value.

In order to initiate the algorithm that determines the optimal gain K by solving the design equations of (16), it is necessary to find an initial gain that stabilizes the system of (3), and if there exist such a K, the system is said to be output stabilizable.

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V. Simulation Results

A detailed simulation study of the proposed controller is carried out. Simulation results show reliable performance and acceptable computation time for real-time implementation. On putting the values of all the constants we get the system matrices as

$$A=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -4.7059 & -0.0882 & 0 & 0 & 1.3588 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -5 & 1.617 & 4.5 \\ 0 & 0 & 0 & 0 & -0.9091 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix},$$

$$B=\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0.8 \end{bmatrix} \quad \text{and} \quad C=\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

The initial gain, at the start of iteration, was assumed to be

$$K_0=\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}$$

The initial conditions for the states were assumed as

$$x_0=\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}^T$$

Also we consider presence of some disturbances d in the system (3) as

$$d=\begin{bmatrix} -0.0043 & -0.0167 & 0.0013 & 0.0029 & -0.0115 & 0.0119 \end{bmatrix}^T$$

Now two cases are taken to illustrate the performance of proposed controller with simulation results.

Case-1: The Q and R matrices were chosen as

$$Q=\begin{bmatrix} 10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and}$$

$$R=100$$

By simulation, after 38 iterations we got the optimal value of controller gain K^* and the performance index J as

$$K^*=\begin{bmatrix} -0.0296 & 0.1650 \\ 0.0339 & 0.2673 \end{bmatrix} \text{ and } J=113.2163$$

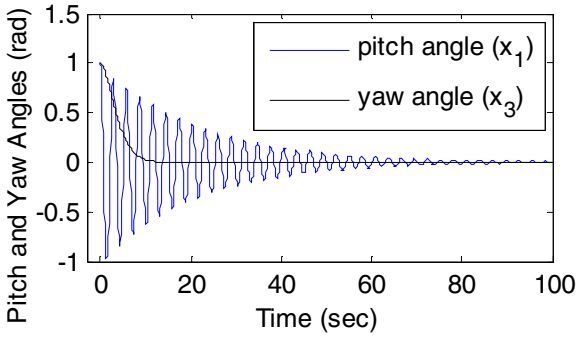


Fig. 2. The outputs (ψ & φ) of TRMS without disturbances

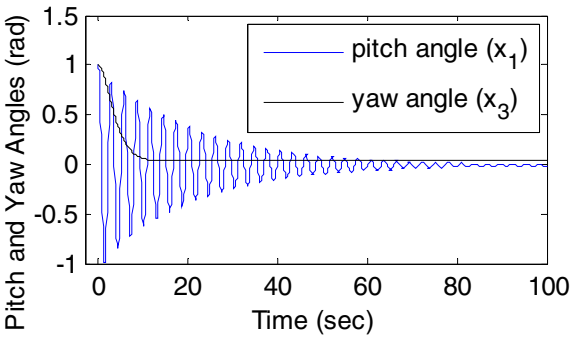


Fig. 3. The outputs (ψ & φ) of TRMS with disturbances

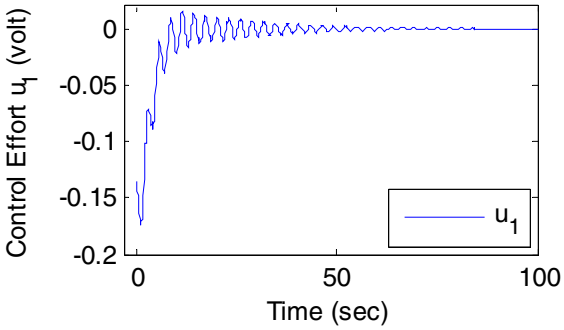


Fig. 4 Control input u_1 to the TRMS

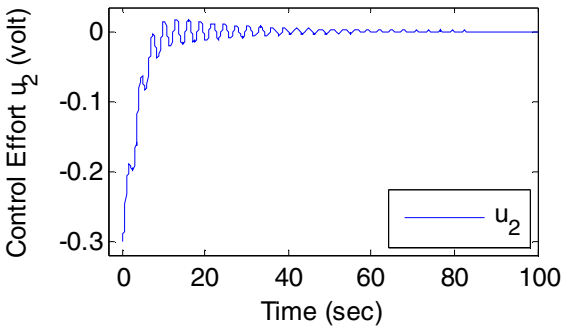


Fig. 5 Control input u_2 to the TRMS

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Case-2: The Q and R matrices were chosen as

$$Q = \begin{bmatrix} 10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ and}$$

$$R = 10$$

By simulation, after 27 iterations we got the optimal value of controller gain K^* and the performance index J as

$$K^* = \begin{bmatrix} -0.2832 & 0.3328 \\ 0.1159 & 0.8174 \end{bmatrix} \text{ and } J = 53.8644$$

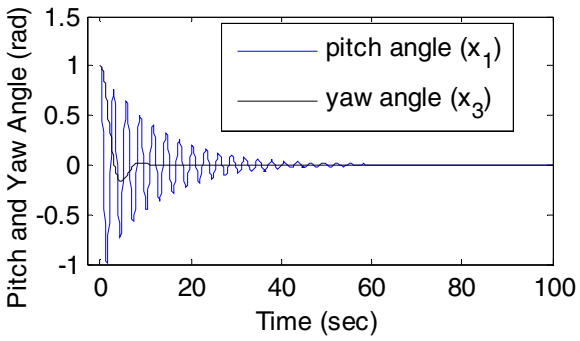


Fig. 6 The outputs (ψ & φ) of TRMS without disturbances

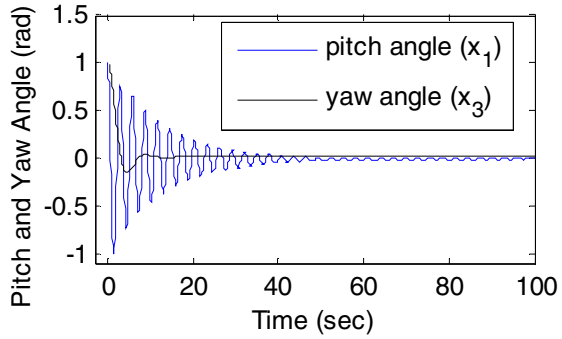


Fig. 7 The outputs (ψ & φ) of TRMS with disturbances

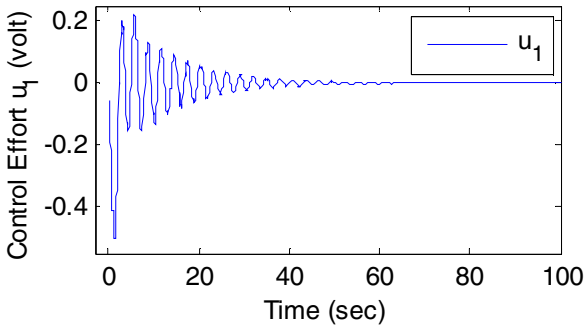


Fig. 8 Control input u_1 to the TRMS

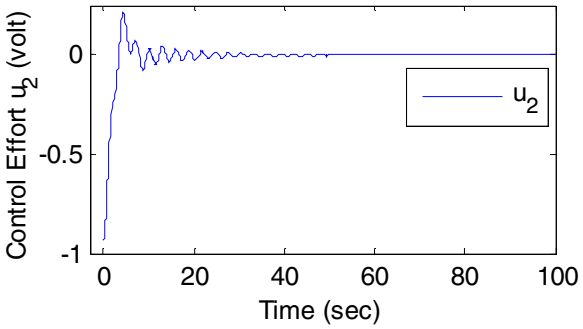


Fig. 9 Control input u_2 to the TRMS

VI. Conclusion

An optimal controller for the twin rotor MIMO system, based on the output feedback technique is presented. Simulation results are shown to present the applicability of the proposed controller. For real time implementation, which is the proposed future work, the experiments have to be carried out on the real-time 2-DOF TRMS system using MATLAB real-time tool box and Advantech PCI1711 card.

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